MDRS SC-32 BAHADDURGHATTA (KOGUNDE)

Chitradurga Taluk & District — 577519

Student Notes

Level 2 — Class 7

ICT Workbook - Level 2

Name: _____

Class: _____ Section: _____

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How to Use This Workbook

Write in pencil or pen. Complete Try It before checking answers. Use Check Yourself to test yourself. Fill in Today I Learned after every lesson. Use the Revision Page before each assessment. The Glossary is at the back.

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Chapter Map — Networking and Internet

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- 1.12 Networking Project

Key terms to know:

Lesson 1.01 Network Types

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can define computer network and explAIn its importance] [- I can identify different types of networks with examples] [- I can compare personal area network, local area network, and wide area network]

Key Words

Network	Group of devices connected to share information and resources together
PAN	Tiny network around one person like phone to watch
LAN	Network covering one building like school lab or office
WAN	Huge network spanning cities or countries like internet globally

Core Idea

A computer network is a group of devices connected together to share data and resources. Networks are classified by size.

Sketch the diagram or concept here.

Worked Example

Step 1: Students identify that inter-school connection is WAN while school lab network is LAN.
Step 2: They draw a simple network diagram showing residential school connections.

Try It (write your answers)

1. In your own words, what is computer network?

Answer: Write a short definition using what you learned today.

2. Give one real-life example of Network Types.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.02 LAN and WAN

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can explAIn the difference between LAN and WAN in detAIl] [- I can describe advantages and disadvantages of each network type] [- I can identify real-world examples of LAN and WAN in Karnataka]

Key Words

LAN	Local network connecting devices inside one building quickly and cheaply
WAN	Wide network connecting distant places through telephone or satellite links
Router	Box directing data between different networks efficiently without delay for speed
Bandwidth	Maximum data that can travel per second across a connection steadily

Core Idea

LAN or Local Area Network connects devices within a small area like a building or campus. It uses Ethernet cables or Wi-Fi.

Sketch the diagram or concept here.

Worked Example

Step 1: Students sketch a campus LAN with labeled devices.
Step 2: They show the Internet connection as a cloud outside the campus boundary.

Try It (write your answers)

1. Explain: Explain the difference between LAN and WAN in detail.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of LAN and WAN.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.03 Internet Services

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can list and explAIn major Internet services avAIlable to users] [- I can differentiate between web browsing, emAIl, and file transfer services] [- I can describe how Internet services help in education at residential schools]

Key Words

Service	Useful facility provided through internet like email or video call
ISP	Company giving internet connection to homes and schools for monthly fee
Protocol	Agreed rule letting different computers talk to each other correctly without confusion
Streaming	Playing video or music directly without downloading whole file completely first

Core Idea

The Internet provides many services. World Wide Web lets you access websites and information. Email lets you send and receive electronic messages.

Sketch the diagram or concept here.

Worked Example

Step 1: Students argue for different services based on educational value.
Step 2: Some choose web search for research, others choose email for communication with teachers, and some choose video for learning.

Try It (write your answers)

1. Explain: List and explain major Internet services available to users.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Internet Services.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

_____

I still need help with:

Lesson 1.04 Email Advanced

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can compose professional emAIls with proper structure and etiquette] [- I can use emAIl features like CC, BCC, and attachments effectively] [- I can understand emAIl security and avoid phishing threats]

Key Words

CC	Sending same email copy to another person visibly
BCC	Sending hidden copy where others cannot see recipient
Reply All	Answering all people included in original email together
Forward	Sending received email to another person with original content

Core Idea

Professional emails have a clear subject line, proper greeting like Dear or respected, concise body paragraphs, and a formal closing with your name an

Sketch the diagram or concept here.

Worked Example

Step 1: students compose complete professional emails with all elements.
Step 2: They verify correct use of CC and BCC and practice attaching files using the school computers.

Try It (write your answers)

1. Explain: Compose professional emails with proper structure and etiquette.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Email Advanced.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.05 Network Topologies

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can define network topology and explAIn its importance] [- I can describe star, bus, ring, and mesh topologies with diagrams] [- I can identify advantages and disadvantages of each topology]

Key Words

Topology	Shape showing how devices are arranged and connected in network physically
Star	All devices connected to one central hub like spokes of wheel
Ring	Devices linked in circle where data travels around sequentially one by one
Bus	All devices sharing one single cable like passengers on same road

Core Idea

Topology is the arrangement of devices in a network. Bus topology connects all devices to one central cable like beads on a string.

Sketch the diagram or concept here.

Worked Example

Step 1: students identify star topology as most common in labs because if one cable fails others continue working.
Step 2: They discuss how central switch makes troubleshooting easy.

Try It (write your answers)

1. In your own words, what is network topology?

Answer: Write a short definition using what you learned today.

2. Give one real-life example of Network Topologies.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.06 TCP/IP Basics

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can explAIn the purpose of TCP and IP in network communication] [- I can describe how data is broken into packets and reassembled] [- I can understand the four-layer TCP/IP model]

Key Words

TCP	Rule ensuring data arrives correctly and in order without missing pieces
IP Address	Unique numbers identifying each device on network like house address
Packet	Small piece of message travelling independently to be reassembled later correctly
Protocol	Set of rules computers follow to communicate without misunderstanding each other

Core Idea

TCP or Transmission Control Protocol breaks data into small packets, numbers them, and ensures all packets arrive correctly at the destination.

Sketch the diagram or concept here.

Worked Example

Step 1: students create a four-layer diagram and place example activities in each layer.
Step 2: They discuss how layers work together like different departments in a company.

Try It (write your answers)

1. Explain: Explain the purpose of TCP and IP in network communication.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of TCP/IP Basics.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

_____

I still need help with:

_____

Lesson 1.07 Network Devices

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify common network devices and their functions] [- I can explAIn the role of routers, switches, and modems in a network] [- I can describe how network devices work together to enable communication]

Key Words

Router	Device connecting different networks and directing traffic between them intelligently for speed
Switch	Box connecting many devices inside one network and forwarding data correctly fast
Hub	Simple device broadcasting data to all connected devices at once together
Access Point	Device letting wireless devices join wired network through Wi-Fi signals freely

Core Idea

A modem converts digital signals from your computer to analog signals for telephone lines and back.

Sketch the diagram or concept here.

Worked Example

Step 1: students propose modem for Internet conversion, router for distribution, switches for connecting computers, and print server for shared printing.
Step 2: They justify placement based on cable length and signal strength.

Try It (write your answers)

1. List or name: common network devices and their functions.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Network Devices.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.08 Wireless Networks

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can explAIn how wireless networks use radio waves to transmit data] [- I can describe Wi-Fi standards and their differences] [- I can set up and secure a basic wireless network connection]

Key Words

Wi-Fi	Radio technology connecting devices without cables over short distance conveniently
Hotspot	Area where Wi-Fi is available for phones and laptops to connect
Signal	Radio wave carrying data through air between devices invisibly around us
Password	Secret word protecting network from unknown users trying to connect

Core Idea

Wireless networks use radio waves to transmit data between devices. Wi-Fi follows IEEE 802.11 standards. 802.11n offers speeds up to six hundred Mbps.

Sketch the diagram or concept here.

Worked Example

Step 1: students walk around campus with devices recording signal bars.
Step 2: They identify dead zones and suggest where additional access points could improve coverage.

Try It (write your answers)

1. Explain: Explain how wireless networks use radio waves to transmit data.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Wireless Networks.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.09 Internet Protocols

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can define Internet protocols and explAIn their purpose] [- I can describe HTTP, HTTPS, FTP, and SMTP protocols with examples] [- I can understand the difference between secure and non-secure protocols]

Key Words

HTTP	Rule for fetching web pages from server to your browser quickly
FTP	Rule for uploading or downloading files between computers efficiently and safely
SMTP	Rule for sending emails from your device to recipient's server reliably
DNS	System translating website names into number addresses automatically for computers

Core Idea

Protocols are rules that govern communication. HTTP or Hypertext Transfer Protocol sends data in plain text without encryption.

Sketch the diagram or concept here.

Worked Example

Step 1: students experience the difference between unencrypted and encrypted communication.
Step 2: The hacker can easily read plain text but cannot decode encrypted messages.

Try It (write your answers)

1. In your own words, what is Internet protocols?

Answer: Write a short definition using what you learned today.

2. Give one real-life example of Internet Protocols.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.10 Network Security

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify common network security threats and attacks] [- I can describe protection methods including passwords, firewalls, and antivirus] [- I can practice safe browsing habits and recognize phishing attempts]

Key Words

Firewall	Barrier blocking unwanted traffic from entering network without permission for safety
Password	Secret code verifying you are allowed to access system or data
Encryption	Scrambling data so only authorised person can read it with key
Virus	Harmful program that copies itself and damages files or steals information

Core Idea

Network security threats include viruses that damage files, worms that spread across networks, phishing that steals passwords through fake sites, and

Sketch the diagram or concept here.

Worked Example

Step 1: students inspect lab computers and find that some have weak passwords like one two three four five six.
Step 2: They recommend password policies and regular updates.

Try It (write your answers)

1. List or name: common network security threats and attacks.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Network Security.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.11 Cloud Services

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can define cloud computing and explAIn its benefits] [- I can identify popular cloud services like Google Drive and OneDrive] [- I can compare local storage versus cloud storage advantages and limitations]

Key Words

Cloud Storage	Saving files online so you access them from anywhere
Backup	Copy of files kept safely to prevent loss from damage
Sync	Keeping same files updated across all your devices automatically
Shared Drive	Folder where many people view and edit same files

Core Idea

Cloud computing means storing and accessing data and programs over the Internet instead of your computer's hard drive.

Sketch the diagram or concept here.

Worked Example

Step 1: students propose benefits like teacher access from anywhere and automatic backup.
Step 2: They identify risks like data breaches and propose encryption and access controls.

Try It (write your answers)

1. In your own words, what is cloud computing?

Answer: Write a short definition using what you learned today.

2. Give one real-life example of Cloud Services.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.12 Networking Project

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can apply networking knowledge to design a complete school network] [- I can present network solutions with proper documentation] [- I can evaluate network designs based on cost, speed, and reliability]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

A network project requires planning several components. Hardware includes computers, servers, switches, routers, and cables.

Sketch the diagram or concept here.

Worked Example

Step 1: teams produce complete network plans with diagrams, equipment lists, and budgets.
Step 2: They present their designs to the class for peer review.

Try It (write your answers)

1. Explain: Apply networking knowledge to design a complete school network.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Networking Project.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Quick Revision — Networking and Internet

1.01 Match each network type to its correct example
1.02 Answer these questions
1.03 Draw a table with columns Service, Purpose, and Example
1.04 Review the email you wrote
1.05 Draw all four network topologies from memory
1.06 Explain in your own words how TCP ensures reliable data transmission
1.07 Match each device to its function
1.08 List three advantages of wireless networks over wired networks
1.09 Match each protocol to its function
1.10 Classify each scenario as safe or risky
1.11 Answer these questions
1.12 Present your network design to the class

Key terms: Network, PAN, LAN, WAN, Router, Bandwidth, Service, ISP, Protocol, Streaming

I still need help with:

Chapter Map — Advanced Graphics

- 2.01 Image Editing Overview
- 2.02 GIMP Interface
- 2.03 Selection Tools
- 2.04 Layers Advanced
- 2.05 Filters and Effects
- 2.06 Color Correction
- 2.07 Text and Typography
- 2.08 Photo Retouching
- 2.09 Compositing
- 2.10 Background Removal
- 2.11 Poster Design
- 2.12 Portfolio Project

Key terms to know:

Lesson 2.01 Image Editing Overview

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand the difference between raster and vector images] [- I can identify common image file formats and their uses] [- I can recognize the importance of image editing in digital communication]

Key Words

Editing	Changing picture to improve look or fix mistakes digitally with tools
Resolution	Number of dots per inch deciding how sharp image looks when printed
Crop	Cutting outer parts to keep only important area of photo
Filter	One-click effect changing colours or brightness of whole picture instantly

Core Idea

Raster images like JPEG and PNG are made of a grid of pixels. Each pixel has a color.

Sketch the diagram or concept here.

Worked Example

Step 1: students categorize images and realize that photos are raster while logos and icons should be vector.
Step 2: They plan to use appropriate formats for their graphics projects.

Try It (write your answers)

1. Explain: Understand the difference between raster and vector images.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Image Editing Overview.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.02 GIMP Interface

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can navigate the GIMP workspace including menus, toolbox, and panels] [- I can customize the GIMP interface for efficient workflow] [- I can create a new image file with specific dimensions and resolution]

Key Words

GIMP	Free program for advanced photo editing like Photoshop without cost
Toolbox	Panel holding brushes selection and paint tools together on side
Canvas	Central area where your picture appears for editing directly with tools
Dock	Side area where layers and brushes panels stay neatly organised together

Core Idea

GIMP or GNU Image Manipulation Program is free image editing software.

Sketch the diagram or concept here.

Worked Example

Step 1: students customize their workspace and learn to reset it if confused.
Step 2: They appreciate having a consistent workspace for efficient editing.

Try It (write your answers)

1. Explain: Navigate the GIMP workspace including menus, toolbox, and panels.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of GIMP Interface.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.03 Selection Tools

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use rectangular, elliptical, and free-form selection tools] [- I can apply selection modifiers like add, subtract, and intersect] [- I can create precise selections to isolate image elements]

Key Words

Selection	Dotted line marking area you want to edit separately from rest
Lasso	Freehand tool for drawing selection around irregular shapes with mouse carefully
Wand	One-click tool selecting areas of similar colour automatically for speed
Feather	Softening edge of selection so change looks natural and blended

Core Idea

Selection tools define which part of an image you want to edit. Rectangle select creates square or rectangular selections.

Sketch the diagram or concept here.

Worked Example

Step 1: students develop precision by combining rectangle select for initial area, ellipse select for head shape, and free select for hair details.
Step 2: They learn zooming helps accuracy.

Try It (write your answers)

1. Explain: Use rectangular, elliptical, and free-form selection tools.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Selection Tools.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.04 Layers Advanced

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand the concept of layers as transparent sheets] [- I can use layer blending modes, opacity, and layer masks] [- I can organize complex projects using layer groups and naming conventions]

Key Words

Layer	Transparent sheet stacked to build complex picture step by step
Background	Bottom layer that holds main colour or photo behind everything
Foreground	Top layer that appears in front of all other layers
Opacity	How see-through or solid a layer appears on screen

Core Idea

Layers are like transparent sheets stacked on top of each other. Each layer can contain different image elements.

Sketch the diagram or concept here.

Worked Example

Step 1: students build multi-layer compositions understanding how each element can be adjusted independently.
Step 2: They discover creative blending mode effects.

Try It (write your answers)

1. Explain: Understand the concept of layers as transparent sheets.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Layers Advanced.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.05 Filters and Effects

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can apply blur, sharpen, and artistic filters to images] [- I can understand when and why to use different filter effects] [- I can combine multiple filters to create unique visual effects]

Key Words

Filter	Ready-made effect that changes photo look with one click
Blur	Softening image so details look less sharp and smooth
Brightness	How light or dark the whole picture appears to eyes
Effect	Special style applied to make picture more attractive quickly

Core Idea

Filters apply mathematical transformations to pixel values.

Sketch the diagram or concept here.

Worked Example

Step 1: students learn that combining filters creates more convincing effects than using one filter alone.
Step 2: They practice building effects progressively.

Try It (write your answers)

1. Explain: Apply blur, sharpen, and artistic filters to images.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Filters and Effects.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.06 Color Correction

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can adjust brightness, contrast, and saturation to improve images] [- I can use color balance to correct color temperature issues] [- I can apply curves and levels for precise tonal adjustments]

Key Words

Brightness	How light or dark the picture appears overall to viewer eyes
Contrast	Difference between darkest and lightest parts of image for clarity
Saturation	Intensity of colours from dull grey to vivid bright appearance
Hue	Name of colour like red blue or green on colour wheel

Core Idea

Color correction fixes problems in photographs. Brightness increases or decreases overall lightness.

Sketch the diagram or concept here.

Worked Example

Step 1: students develop judgment about which corrections to apply and in what order.
Step 2: They learn that sometimes multiple adjustments are needed for the best result.

Try It (write your answers)

1. Explain: Adjust brightness, contrast, and saturation to improve images.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Color Correction.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.07 Text and Typography

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can add and format text with fonts, sizes, and colors in GIMP] [- I can understand basic typography principles for readable design] [- I can create text effects using layers and filters]

Key Words

Font	Design style of letters like Arial or Noto Sans for reading
Size	How big or small the text appears on page or screen
Alignment	Positioning text to left centre or right of page neatly
Typography	Art of arranging text to look beautiful and readable together

Core Idea

Typography is the art of arranging text for readability and visual appeal. Font choice affects mood. Serif fonts like Times feel formal.

Sketch the diagram or concept here.

Worked Example

Step 1: students apply typography principles to create effective poster designs.
Step 2: They learn that text effects should enhance readability not reduce it.

Try It (write your answers)

1. Explain: Add and format text with fonts, sizes, and colors in GIMP.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Text and Typography.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.08 Photo Retouching

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can remove blemishes and unwanted objects from photographs] [- I can use clone stamp and healing tools effectively] [- I can retouch portrAIts while mAIntAIning natural appearance]

Key Words

Retouch	Fixing small flaws like spots or red eyes in photo digitally
Clone	Copying clean area to cover unwanted spot seamlessly without trace
Heal	Blending copied area so repair looks natural and invisible to eyes
Patch	Replacing damaged part with good texture from nearby area smoothly

Core Idea

Photo retouching improves images by removing imperfections. The heal tool samples texture from surrounding areas and blends it to cover blemishes.

Sketch the diagram or concept here.

Worked Example

Step 1: students practice subtle retouching that enhances without obvious manipulation.
Step 2: They understand that the best retouching is unnoticeable.

Try It (write your answers)

1. Explain: Remove blemishes and unwanted objects from photographs.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Photo Retouching.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.09 Compositing

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can combine elements from multiple images into a single composition] [- I can match color and lighting between different source images] [- I can create seamless blends using layer masks and blending modes]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Compositing combines elements from different images into one realistic scene.

Sketch the diagram or concept here.

Worked Example

Step 1: students create imaginative composites while applying technical skills.
Step 2: They learn that patience and attention to detail create convincing results.

Try It (write your answers)

1. Explain: Combine elements from multiple images into a single composition.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Compositing.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.10 Background Removal

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can remove backgrounds from photographs to isolate subjects] [- I can use selection tools and layer masks for clean extraction] [- I can create transparent backgrounds for design projects]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Background removal isolates the subject from its surroundings.

Sketch the diagram or concept here.

Worked Example

Step 1: students complete a practical background removal project.
Step 2: They learn to output files in appropriate formats for different uses.

Try It (write your answers)

1. Explain: Remove backgrounds from photographs to isolate subjects.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Background Removal.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.11 Poster Design

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can apply design principles to create effective educational posters] [- I can use hierarchy, alignment, and contrast for visual communication] [- I can design posters for school events with clear messaging]

Key Words

Poster	Large design combining text and images to share message
Layout	Arrangement of text and pictures to look balanced neatly
Heading	Big bold words at top that catch attention quickly
Alignment	Lining up text and images neatly along invisible lines

Core Idea

Poster design follows key principles. Hierarchy guides the viewer eye from most to least important elements using size and placement.

Sketch the diagram or concept here.

Worked Example

Step 1: students create complete event posters applying all design principles.
Step 2: They practice planning before designing and iterating based on feedback.

Try It (write your answers)

1. Explain: Apply design principles to create effective educational posters.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Poster Design.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.12 Portfolio Project

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can compile all graphics skills into a cohesive portfolio] [- I can present digital work with proper documentation] [- I can reflect on learning progress and skill development]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

A portfolio is a curated collection of your best work that demonstrates your skills and growth.

Sketch the diagram or concept here.

Worked Example

Step 1: students compile professional portfolios demonstrating their complete skill set.
Step 2: They practice file management and presentation skills.

Try It (write your answers)

1. Explain: Compile all graphics skills into a cohesive portfolio.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Portfolio Project.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Quick Revision — Advanced Graphics

2.01 Answer these questions
2.02 Identify each GIMP interface element
2.03 Complete these tasks
2.04 Describe what layers are and why they are useful
2.05 Apply Gaussian Blur to a background layer while keeping the subject sharp
2.06 Correct this overexposed photograph using appropriate tools
2.07 Create a text composition with at least three different fonts that work…
2.08 Retouch this portrait to remove three blemishes while keeping the natural skin…
2.09 Evaluate this composite image for realism
2.10 Compare three background removal methods on the same image
2.11 Evaluate your poster using the design principles checklist
2.12 Present your portfolio to the class

Key terms: Editing, Resolution, Crop, Filter, GIMP, Toolbox, Canvas, Dock, Selection, Lasso

I still need help with:

Chapter Map — Multimedia

- 3.01 What is Multimedia
- 3.02 Audio Editing Audacity
- 3.03 Sound Recording
- 3.04 Audio Effects
- 3.05 Video Editing Basics
- 3.06 Video Transitions
- 3.07 Animation Principles
- 3.08 Presentation Software
- 3.09 Interactive Presentations
- 3.10 Digital Storytelling
- 3.11 Podcasting
- 3.12 Multimedia Project

Key terms to know:

Lesson 3.01 What is Multimedia

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can define multimedia and identify its components] [- I can understand how text, audio, video, and graphics work together] [- I can recognize multimedia applications in dAIly life]

Key Words

Multimedia	Combination of text sound images and video together to communicate
Media	Single form like audio or video used inside larger presentation
Integration	Combining different media types into one smooth presentation for audience
Interactive	Letting viewer click or choose what happens next inside project

Core Idea

Multimedia means multiple forms of media working together.

Sketch the diagram or concept here.

Worked Example

Step 1: students analyze apps identifying text labels, graphic characters, sound effects, background music, and animated transitions.
Step 2: They see how elements combine for engagement.

Try It (write your answers)

1. In your own words, what is multimedia?

Answer: Write a short definition using what you learned today.

2. Give one real-life example of Multimedia.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.02 Audio Editing Audacity

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can navigate the Audacity interface and understand its mAIn features] [- I can import, record, and export audio files in different formats] [- I can apply basic editing operations like cut, copy, and paste to audio]

Key Words

Audio	Sound or music recorded and played through computer speakers
Record	Saving your voice or sound as a digital file
Playback	Listening again to recorded sound through speakers clearly
Microphone	Device that captures your voice and sends to computer

Core Idea

Audacity is free audio editing software. The interface has the transport controls at top left for record, play, stop, and rewind.

Sketch the diagram or concept here.

Worked Example

Step 1: students complete a practical editing project.
Step 2: They practice selection precision and learn to maintain audio quality during editing.

Try It (write your answers)

1. Explain: Navigate the Audacity interface and understand its main features.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Audio Editing Audacity.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.03 Sound Recording

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can set up microphone and recording environment for quality audio] [- I can understand audio settings like sample rate and bit depth] [- I can record clear voice audio with proper levels and techniques]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Good audio recording requires attention to environment, equipment, and settings. Environment should be quiet with soft surfaces to reduce echo.

Sketch the diagram or concept here.

Worked Example

Step 1: students apply recording techniques in a practical setting.
Step 2: They learn that preparation and environment are as important as equipment.

Try It (write your answers)

1. Explain: Set up microphone and recording environment for quality audio.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Sound Recording.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.04 Audio Effects

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can apply noise reduction to clean up audio recordings] [- I can use equalization to adjust frequency balance] [- I can add effects like reverb and compression for professional sound]

Key Words

Audio	Sound or music recorded and played through computer speakers
Record	Saving your voice or sound as a digital file
Playback	Listening again to recorded sound through speakers clearly
Microphone	Device that captures your voice and sends to computer

Core Idea

Audio effects improve sound quality. Noise reduction removes background hiss and hum by sampling noise and subtracting it.

Sketch the diagram or concept here.

Worked Example

Step 1: students apply a complete effects chain transforming amateur recordings into professional-sounding audio.
Step 2: They document their settings for each effect.

Try It (write your answers)

1. Explain: Apply noise reduction to clean up audio recordings.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Audio Effects.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.05 Video Editing Basics

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can import video clips and organize them in a video editor] [- I can trim, split, and arrange clips on a timeline] [- I can add titles and basic transitions between clips]

Key Words

Video	Moving pictures made from many still images played quickly
Frame	Single still picture that is part of a longer video
Timeline	Bar showing order and length of clips in your project
Clip	Short piece of video you can move or trim separately

Core Idea

Video editing arranges clips to tell a story. Import brings media files into the project. The timeline shows clips from left to right in playback order.

Sketch the diagram or concept here.

Worked Example

Step 1: students complete a practical video project.
Step 2: They practice making editorial decisions about what to keep and what to cut.

Try It (write your answers)

1. Explain: Import video clips and organize them in a video editor.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Video Editing Basics.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.06 Video Transitions

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify different types of video transitions and their uses] [- I can apply transitions appropriately based on content mood] [- I can avoid overuse of transitions that distract from content]

Key Words

Video	Moving pictures made from many still images played quickly
Frame	Single still picture that is part of a longer video
Timeline	Bar showing order and length of clips in your project
Clip	Short piece of video you can move or trim separately

Core Idea

Transitions connect scenes and control pacing. Cut is instant and most common used for same-time same-place scene changes.

Sketch the diagram or concept here.

Worked Example

Step 1: students demonstrate understanding of transition purpose.
Step 2: They can justify each transition choice with content-based reasoning.

Try It (write your answers)

1. List or name: different types of video transitions and their uses.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Video Transitions.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.07 Animation Principles

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand the twelve principles of animation] [- I can create simple animations using keyframes] [- I can apply timing and spacing for realistic motion]

Key Words

Animation	Creating movement by showing many pictures very quickly together
Frame	One picture in sequence that creates motion when played fast
Motion	Change of position of object from one place another
Timeline	Line showing order and timing of each frame in animation

Core Idea

Animation principles guide believable motion. Squash and stretch gives weight and flexibility. Anticipation prepares the viewer for an action.

Sketch the diagram or concept here.

Worked Example

Step 1: students apply multiple principles to create complex believable motion.
Step 2: They learn that combining principles produces better results than using one alone.

Try It (write your answers)

1. Explain: Understand the twelve principles of animation.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Animation Principles.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.08 Presentation Software

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can create effective slide presentations with appropriate content density] [- I can use themes, layouts, and master slides for consistent design] [- I can incorporate multimedia elements into presentations]

Key Words

Slide	Single page inside presentation showing text and images together
Transition	Effect that moves you smoothly from one slide next
Animation	Movement added to text or pictures on a slide
Template	Ready-made design that keeps all slides looking consistent neatly

Core Idea

Effective presentations follow the ten-twenty-thirty rule. Ten slides maximum for a twenty-minute presentation with thirty point minimum font.

Sketch the diagram or concept here.

Worked Example

Step 1: students create complete professional presentations.
Step 2: They practice balancing visual content with verbal explanation.

Try It (write your answers)

1. Explain: Create effective slide presentations with appropriate content density.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Presentation Software.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.09 Interactive Presentations

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can add hyperlinks and navigation to create non-linear presentations] [- I can embed quizzes and interactive elements for audience engagement] [- I can design presentations that adapt based on audience choices]

Key Words

Slide	Single page inside presentation showing text and images together
Transition	Effect that moves you smoothly from one slide next
Animation	Movement added to text or pictures on a slide
Template	Ready-made design that keeps all slides looking consistent neatly

Core Idea

Interactive presentations break the linear slide-by-slide format. Hyperlinks connect text or objects to specific slides creating navigation options.

Sketch the diagram or concept here.

Worked Example

Step 1: students build complete interactive quiz experiences.
Step 2: They practice designing clear navigation and meaningful feedback for each answer.

Try It (write your answers)

1. Explain: Add hyperlinks and navigation to create non-linear presentations.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Interactive Presentations.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.10 Digital Storytelling

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can structure a narrative using beginning, middle, and end] [- I can combine text, images, audio, and video for effective storytelling] [- I can create emotional impact through multimedia story elements]

Key Words

Story	Sequence of events told with words pictures and sound together
Narrative	Way you organise story with beginning middle and end
Scene	Single part of story showing one moment or place
Voiceover	Recorded speech added over pictures to explain story clearly

Core Idea

Digital storytelling uses multimedia to tell narratives. Structure follows the story arc. Beginning introduces characters and setting.

Sketch the diagram or concept here.

Worked Example

Step 1: students produce complete digital stories integrating all multimedia skills.
Step 2: They practice storytelling craft alongside technical skills.

Try It (write your answers)

1. Explain: Structure a narrative using beginning, middle, and end.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Digital Storytelling.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.11 Podcasting

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can plan and structure a podcast episode with segments] [- I can record and edit multi-segment audio programs] [- I can add intro music, transitions, and outro to podcast episodes]

Key Words

Podcast	Series of audio episodes you can listen anytime online
Episode	Single audio story that is part of larger series
Host	Person who speaks and guides the podcast for listeners
Subscribe	Following podcast to receive new episodes automatically when released

Core Idea

A podcast is a regularly released audio program with episodes.

Sketch the diagram or concept here.

Worked Example

Step 1: students produce complete podcast episodes.
Step 2: They practice interviewing, segment transitions, and maintaining consistent audio quality throughout.

Try It (write your answers)

1. Explain: Plan and structure a podcast episode with segments.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Podcasting.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.12 Multimedia Project

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can plan and execute a complete multimedia project] [- I can integrate multiple media types into a cohesive final product] [- I can present multimedia work with professional documentation]

Key Words

Multimedia	Combination of text sound images and video together to communicate
Media	Single form like audio or video used inside larger presentation
Integration	Combining different media types into one smooth presentation for audience
Interactive	Letting viewer click or choose what happens next inside project

Core Idea

A multimedia project requires careful planning. Define your topic and audience. Create a project proposal with objectives and media plan.

Sketch the diagram or concept here.

Worked Example

Step 1: students produce complete multimedia projects demonstrating integrated skills.
Step 2: They practice project management and documentation.

Try It (write your answers)

1. Explain: Plan and execute a complete multimedia project.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Multimedia Project.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Quick Revision — Multimedia

3.01 Classify each example as multimedia or single media
3.02 Complete these tasks
3.03 Record the same paragraph three times with different microphone positions
3.04 Apply noise reduction to this recording and explain your settings
3.05 Trim this thirty-second clip to remove the first five seconds and last three…
3.06 Identify the transition type used between each pair of clips in this sample…
3.07 Identify which animation principles are used in this sample animation
3.08 Evaluate this presentation against the ten-twenty-thirty rule
3.09 Test your interactive presentation with a classmate
3.10 Present your digital story to the class
3.11 Listen to your podcast and evaluate it against these criteria
3.12 Present your multimedia project to the class

Key terms: Multimedia, Media, Integration, Interactive, Audio, Record, Playback, Microphone, Concept, Example

I still need help with:

Chapter Map — Data Handling

- 4.01 Spreadsheet Overview
- 4.02 Data Entry and Formatting
- 4.03 Basic Formulas
- 4.04 Functions SUM AVERAGE
- 4.05 Charts and Graphs
- 4.06 Sorting Data
- 4.07 Filtering Data
- 4.08 Conditional Formatting
- 4.09 Data Validation
- 4.10 Pivot Tables Intro
- 4.11 Data Analysis
- 4.12 Budget Project

Key terms to know:

Lesson 4.01 Spreadsheet Overview

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify the purpose and benefits of spreadsheet software] [- I can navigate the spreadsheet interface including cells, rows, and columns] [- I can enter and edit data in spreadsheet cells]

Key Words

Cell	Single box at intersection of row and column holding data
Range	Group of selected cells like A1 to C5 for calculation together
Formula	Equation starting with equals calculating result from other cells automatically
Worksheet	Single page of spreadsheet containing grid of cells for work

Core Idea

A spreadsheet is a grid of cells organized in rows numbered one, two, three and columns labeled A, B, C. Each cell holds one piece of data.

Sketch the diagram or concept here.

Worked Example

Step 1: students apply spreadsheet basics to create a practical document.
Step 2: They learn formatting improves readability of data tables.

Try It (write your answers)

1. List or name: the purpose and benefits of spreadsheet software.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Spreadsheet Overview.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

_____

I still need help with:

_____

Lesson 4.02 Data Entry and Formatting

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can enter different data types including text, numbers, and dates] [- I can apply cell formatting for alignment, borders, and colors] [- I can use fill handle and auto-fill for efficient data entry]

Key Words

Data	Facts like numbers and words stored on a computer
Raw Data	Unprocessed facts collected before cleaning or organising properly
Information	Useful meaning produced after processing raw data carefully
Dataset	Collection of related data stored together in one place

Core Idea

Data types determine how information is stored and displayed. Text aligns left by default. Numbers align right.

Sketch the diagram or concept here.

Worked Example

Step 1: students create professionally formatted spreadsheets.
Step 2: They practice combining multiple formatting features for clear data presentation.

Try It (write your answers)

1. Explain: Enter different data types including text, numbers, and dates.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Data Entry and Formatting.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.03 Basic Formulas

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can write formulas using cell references for automatic calculations] [- I can use arithmetic operators like add, subtract, multiply, and divide] [- I can understand the order of operations in spreadsheet formulas]

Key Words

Formula	Math expression starting with equals that calculates answer automatically for you
Operator	Symbol like plus or minus telling what math to do
Reference	Address like B2 pointing to value in another cell for calculation
Equals	Sign at formula start showing calculation should happen immediately after typing

Core Idea

Formulas start with equals sign and use arithmetic operators.

Sketch the diagram or concept here.

Worked Example

Step 1: students apply formulas to real-world calculation scenarios.
Step 2: They see how one change cascades through all dependent formulas.

Try It (write your answers)

1. Explain: Write formulas using cell references for automatic calculations.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Basic Formulas.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.04 Functions SUM AVERAGE

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use built-in functions for common calculations] [- I can apply SUM to add ranges of numbers and AVERAGE to find mean] [- I can understand function syntax with arguments and ranges]

Key Words

SUM	Function adding all numbers in selected range quickly without manual typing
AVERAGE	Function dividing total by count to find middle value for group
MAX	Function finding largest number in selected range instantly for comparison
MIN	Function finding smallest number in selected range instantly for comparison

Core Idea

Functions are pre-built formulas that perform common calculations. Syntax is function name followed by parentheses containing arguments.

Sketch the diagram or concept here.

Worked Example

Step 1: students apply multiple functions to create a comprehensive grade analysis.
Step 2: They understand how functions work together for complete data analysis.

Try It (write your answers)

1. Explain: Use built-in functions for common calculations.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Functions SUM AVERAGE.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.05 Charts and Graphs

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can create bar charts, line graphs, and pie charts from spreadsheet data] [- I can choose the appropriate chart type for different data] [- I can customize chart titles, labels, and legends for clarity]

Key Words

Chart	Picture like bar or pie showing data visually for understanding
Axis	Line along side or bottom showing measurement scale clearly
Legend	Key explaining what each colour or symbol means in chart
Data Point	Single value plotted as dot or bar inside chart

Core Idea

Charts visualize data for better understanding. Bar charts compare quantities across categories using rectangular bars.

Sketch the diagram or concept here.

Worked Example

Step 1: students create multiple chart types from the same data.
Step 2: They practice customizing chart elements for professional presentation.

Try It (write your answers)

1. Explain: Create bar charts, line graphs, and pie charts from spreadsheet data.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Charts and Graphs.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.06 Sorting Data

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can sort spreadsheet data in ascending and descending order] [- I can sort by multiple columns for complex data organization] [- I can understand how sorting affects row relationships]

Key Words

Sort	Putting rows in alphabetical or numerical order like A-Z quickly
Ascending	Order from smallest to largest or A to Z upwards progressively
Descending	Order from largest to smallest or Z to A downwards progressively
Key	Column chosen as basis for deciding sorted order of rows

Core Idea

Sorting rearranges rows based on values in one or more columns. Ascending order puts smallest values first. Descending order puts largest values first.

Sketch the diagram or concept here.

Worked Example

Step 1: students practice multi-level sorting for complex data analysis.
Step 2: They discover that primary and secondary sort keys create meaningful groupings.

Try It (write your answers)

1. Explain: Sort spreadsheet data in ascending and descending order.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Sorting Data.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.07 Filtering Data

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can apply filters to dISPlay only specific rows matching criteria] [- I can use multiple filter conditions for complex data queries] [- I can clear filters and understand their effect on data view]

Key Words

Filter	Hiding rows that do not meet condition and showing only matches
Criteria	Rule deciding which rows stay visible after filtering operation for accuracy
Dropdown	Small menu appearing when you click filter arrow for choices
Clear Filter	Removing filter to show all rows again as before

Core Idea

Filtering hides rows that do not match your criteria showing only relevant data. AutoFilter adds dropdown arrows to headers.

Sketch the diagram or concept here.

Worked Example

Step 1: students practice layered filtering for complex data queries.
Step 2: They understand how combining filters narrows results progressively.

Try It (write your answers)

1. Explain: Apply filters to display only specific rows matching criteria.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Filtering Data.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.08 Conditional Formatting

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can apply color scales to visualize data values with colors] [- I can create data bars that show magnitude visually] [- I can set icon sets to indicate value thresholds]

Key Words

Conditional Formatting	Automatically colouring cells that meet a rule like marks below 35
Rule	Condition like greater than 80 that decides which cells change colour
Highlight	Colouring cell background to draw attention to important values quickly
Scale	Graduated colours showing low to high values across range visually nicely

Core Idea

Conditional formatting changes cell appearance based on their values. Color scales apply gradient colors from low to high values.

Sketch the diagram or concept here.

Worked Example

Step 1: students apply multiple conditional formatting rules to create informative data visualizations.
Step 2: They see how formatting reveals data stories.

Try It (write your answers)

1. Explain: Apply color scales to visualize data values with colors.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Conditional Formatting.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.09 Data Validation

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can set rules that control what data can be entered in cells] [- I can create dropdown lists for consistent data entry] [- I can dISPlay input messages and error alerts for data quality]

Key Words

Validation	Checking entered data matches allowed type or range before accepting
Rule	Limit like only numbers between 1 and 100 allowed in cell
Error Alert	Message warning you entered wrong type of data for cell
Dropdown List	Menu limiting choice to approved options like Yes or No

Core Idea

Data validation controls what users can enter in cells.

Sketch the diagram or concept here.

Worked Example

Step 1: students build complete validated data entry systems.
Step 2: They understand that validation creates reliable data for analysis.

Try It (write your answers)

1. Explain: Set rules that control what data can be entered in cells.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Data Validation.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.10 Pivot Tables Intro

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand what pivot tables do and when to use them] [- I can create a basic pivot table from a data set] [- I can summarize data using rows columns and values]

Key Words

Pivot Table	Summarising large data quickly by grouping and counting with drag
Row Label	Category appearing along left side of pivot table for grouping
Value	Number being counted summed or averaged inside pivot table results
Filter	Selecting which data to include or exclude from summary quickly

Core Idea

Pivot tables summarize large datasets into meaningful reports. They have four areas. Rows define what to group by like school names.

Sketch the diagram or concept here.

Worked Example

Step 1: students create multi-dimensional summaries using pivot tables.
Step 2: They discover how quickly pivot tables answer complex data questions.

Try It (write your answers)

1. Explain: Understand what pivot tables do and when to use them.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Pivot Tables Intro.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.11 Data Analysis

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use multiple spreadsheet tools together for comprehensive analysis] [- I can interpret data patterns and draw meaningful conclusions] [- I can present data findings with charts and written analysis]

Key Words

Data	Facts like numbers and words stored on a computer
Raw Data	Unprocessed facts collected before cleaning or organising properly
Information	Useful meaning produced after processing raw data carefully
Dataset	Collection of related data stored together in one place

Core Idea

Data analysis combines multiple techniques. First organize data with proper formatting and validation. Then sort to see rankings.

Sketch the diagram or concept here.

Worked Example

Step 1: students complete comprehensive analysis projects.
Step 2: They practice data interpretation and report writing for real-world application.

Try It (write your answers)

1. Explain: Use multiple spreadsheet tools together for comprehensive analysis.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Data Analysis.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.12 Budget Project

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can create a complete budget spreadsheet with income and expenses] [- I can apply formulas for totals, balances, and percentage calculations] [- I can use charts to visualize budget allocation and spending patterns]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

A budget tracks income and expenses over a period.

Sketch the diagram or concept here.

Worked Example

Step 1: students create functional budget tools.
Step 2: They practice realistic financial planning and data visualization for personal finance management.

Try It (write your answers)

1. Explain: Create a complete budget spreadsheet with income and expenses.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Budget Project.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Quick Revision — Data Handling

4.01 Answer these questions
4.02 Format this raw data table with proper alignment, borders, colors, and number…
4.03 Write a formula to calculate the area of a rectangle using cell references for…
4.04 Write a SUM function to add cells B2 through B20
4.05 Choose the best chart type to show student marks distribution
4.06 Sort this data by student name alphabetically
4.07 Apply a filter to show only students from Bengaluru with marks between sixty…
4.08 Apply a color scale to this temperature data from red for cold to blue for hot
4.09 Set up validation to accept only dates in the current school year
4.10 Create a pivot table showing total marks per student
4.11 Present your data analysis findings to the class
4.12 Present your budget spreadsheet

Key terms: Cell, Range, Formula, Worksheet, Data, Raw Data, Information, Dataset, Operator, Reference

I still need help with:

Chapter Map — Programming with Scratch

- 5.01 Scratch Review
- 5.02 Variables Advanced
- 5.03 Conditions If-Else
- 5.04 Nested Conditions
- 5.05 Loops Advanced
- 5.06 Lists
- 5.07 Clones
- 5.08 Pen Extensions
- 5.09 Music Extension
- 5.10 Game Mechanics
- 5.11 AI and ML Intro
- 5.12 Capstone Game

Key terms to know:

Lesson 5.01 Scratch Review

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can recall Scratch interface elements and basic block categories] [- I can review event blocks, motion blocks, and control blocks] [- I can rebuild a simple project from memory to reinforce fundamentals]

Key Words

Scratch	Visual programming tool using blocks to create stories and games
Stage	Area where your characters move and show their actions live
Sprite	Character or object that moves and acts on the stage
Block	Colourful puzzle piece representing one programming instruction visually

Core Idea

Scratch has a stage where the program runs, sprites that perform actions, and scripts that control sprite behavior.

Sketch the diagram or concept here.

Worked Example

Step 1: Students rebuild projects using accumulated knowledge.
Step 2: They practice combining blocks from different categories for richer interactions.

Try It (write your answers)

1. Explain: Recall Scratch interface elements and basic block categories.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Scratch Review.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.02 Variables Advanced

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can create and use variables for storing and updating values] [- I can understand local versus global variable scope] [- I can apply variables in calculations and program logic]

Key Words

Variable	Named container storing value that program can change later dynamically
Scope	Part of program where variable can be seen and used correctly
Assignment	Storing a value into variable using equals sign for later use
Data Type	Kind of value like number or text deciding what operations allowed

Core Idea

Variables are named containers that store data values. Create a variable with the Make a Variable button. Set block assigns an initial value.

Sketch the diagram or concept here.

Worked Example

Step 1: Students build practical variable applications.
Step 2: They learn that variables enable user interaction and dynamic program state.

Try It (write your answers)

1. Explain: Create and use variables for storing and updating values.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Variables Advanced.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.03 Conditions If-Else

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use if-then blocks to make decisions in programs] [- I can apply if-then-else blocks for binary choices] [- I can combine conditions with comparison and logical operators]

Key Words

If	Decision checking condition and running code only when true otherwise skipping
Else	Alternative block running when if condition proves false for completeness
Elif	Extra check when first condition fails but another might succeed sequentially
Condition	Question answerable as true or false deciding which path program takes

Core Idea

If-then blocks check a condition and execute the enclosed blocks only if the condition is true. If-then-else blocks provide two paths.

Sketch the diagram or concept here.

Worked Example

Step 1: Students build interactive programs with conditional logic.
Step 2: They experience how conditions create branching program behavior.

Try It (write your answers)

1. Explain: Use if-then blocks to make decisions in programs.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Conditions If-Else.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.04 Nested Conditions

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can create programs with conditions inside other conditions] [- I can understand the order of evaluation in nested if blocks] [- I can apply nested conditions for complex decision scenarios]

Key Words

Nested If	If statement placed inside another if for more detailed decision making
Branch	Separate path program can take depending on different condition results possible
Logic	Reasoning combining many conditions using and or or correctly together
Flow	Order in which program checks conditions from outer to inner sequentially

Core Idea

Nested conditions place one if block inside another. The outer condition is checked first. If it is false the inner condition is never checked.

Sketch the diagram or concept here.

Worked Example

Step 1: Students build multi-level decision programs.
Step 2: They practice organizing complex logic in readable nested structures.

Try It (write your answers)

1. Explain: Create programs with conditions inside other conditions.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Nested Conditions.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.05 Loops Advanced

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use repeat-until loops for condition-based repetition] [- I can combine loops with conditions for complex behavior] [- I can avoid infinite loops by ensuring termination conditions]

Key Words

Loop	Structure repeating block of code many times without rewriting same lines
For Loop	Repeating fixed number of times using counter variable automatically for control
While Loop	Repeating while condition stays true checking each time before running
Break	Command stopping loop immediately even if condition still true for exit

Core Idea

Repeat-until loops execute blocks repeatedly until a condition becomes true. The condition is checked before each iteration.

Sketch the diagram or concept here.

Worked Example

Step 1: Students build interactive games using repeat-until loops.
Step 2: They experience how loops create continuous gameplay that ends at a specific condition.

Try It (write your answers)

1. Explain: Use repeat-until loops for condition-based repetition.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Loops Advanced.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.06 Lists

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can create and manipulate lists to store multiple values] [- I can use list operations like add, delete, and insert] [- I can apply loops to process all items in a list]

Key Words

List	Ordered collection storing many values together under one name like marks
Index	Position number starting at zero showing where item sits in list
Element	Single value stored at one position inside a list for retrieval
Append	Adding new item to end of existing list quickly without rebuilding

Core Idea

Lists store ordered collections of values. Create a list with Make a List button. Add block appends an item to the end.

Sketch the diagram or concept here.

Worked Example

Step 1: Students build practical list-based applications.
Step 2: They experience how lists enable dynamic data management in programs.

Try It (write your answers)

1. Explain: Create and manipulate lists to store multiple values.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Lists.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.07 Clones

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can create clones of sprites for dynamic program elements] [- I can use clone-specific blocks for individual clone behavior] [- I can manage clone lifecycle from creation to deletion]

Key Words

Clone	Duplicate copy of sprite that acts independently from original on stage
Create Clone	Making new copy of sprite during program run with code
Delete Clone	Removing extra copies when no longer needed to save memory
Control	Block deciding when and how many clones appear on stage

Core Idea

Clones are temporary copies of a sprite created during program execution. Use create clone of block to make a new clone.

Sketch the diagram or concept here.

Worked Example

Step 1: Students build particle simulation effects using clones.
Step 2: They experience how clones create complex visual effects efficiently.

Try It (write your answers)

1. Explain: Create clones of sprites for dynamic program elements.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Clones.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.08 Pen Extensions

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use the pen extension to draw graphics programmatically] [- I can control pen color, size, and transparency for visual effects] [- I can create geometric patterns and art using loop and pen blocks]

Key Words

Pen	Extension allowing sprite to draw lines as it moves across stage
Stamp	Copying sprite image permanently onto stage background like printing once
Clear	Removing all pen drawings from stage to start fresh again
Extension	Extra feature adding new blocks like pen or music to Scratch

Core Idea

The pen extension lets sprites draw on the stage. Pen down starts drawing a trail as the sprite moves. Pen up stops drawing.

Sketch the diagram or concept here.

Worked Example

Step 1: Students create complex generative art using nested loops and pen commands.
Step 2: They discover mathematical beauty in programming.

Try It (write your answers)

1. Explain: Use the pen extension to draw graphics programmatically.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Pen Extensions.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.09 Music Extension

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can add musical notes and rhythms to Scratch projects] [- I can create melodies by sequencing note blocks] [- I can combine music with animation for multimedia projects]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

The music extension provides blocks for creating audio. Play note block specifies pitch using note numbers where sixty is middle C.

Sketch the diagram or concept here.

Worked Example

Step 1: Students create complete musical performances combining audio and visual elements.
Step 2: They practice planning before programming.

Try It (write your answers)

1. Explain: Add musical notes and rhythms to Scratch projects.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Music Extension.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.10 Game Mechanics

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can design core game mechanics including scoring and lives] [- I can implement collision detection for interactive gameplay] [- I can create game loops that manage start, play, and end states]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Game mechanics are rules that control gameplay. Scoring uses a variable that changes based on player actions.

Sketch the diagram or concept here.

Worked Example

Step 1: Students implement complete game mechanics.
Step 2: They practice combining movement, collision, scoring, and game state management.

Try It (write your answers)

1. Explain: Design core game mechanics including scoring and lives.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Game Mechanics.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.11 AI and ML Intro

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand the basic concepts of artificial intelligence and machine learning] [- I can explore how AI recognizes patterns in data] [- I can create a simple Scratch project that simulates AI decision making]

Key Words

AI	Machines that learn and make decisions like humans without explicit programming rules
Machine Learning	Training computers by showing many examples instead of writing rules
Training	Feeding data to model so it learns patterns and improves accuracy gradually
Prediction	Guess made by trained model about new unseen data based on learning

Core Idea

Artificial intelligence or AI is computer software that mimics human thinking.

Sketch the diagram or concept here.

Worked Example

Step 1: Students build rule-based systems that simulate AI reasoning.
Step 2: They understand that simple AI follows decision trees based on user input.

Try It (write your answers)

1. Explain: Understand the basic concepts of artificial intelligence and machine learning.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of AI and ML Intro.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.12 Capstone Game

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can apply all Scratch skills to create a complete game] [- I can plan, design, build, and test a game project] [- I can present game with documentation and demonstration]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

A capstone project demonstrates your complete learning.

Sketch the diagram or concept here.

Worked Example

Step 1: Students build complete games applying all learned concepts.
Step 2: They practice project management and iterative development.

Try It (write your answers)

1. Explain: Apply all Scratch skills to create a complete game.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Capstone Game.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Quick Revision — Programming with Scratch

5.01 List all eight block categories with one block from each
5.02 Create a variable called timer that counts down from sixty
5.03 Write a program that makes a sprite grow big if the space key is pressed and…
5.04 Trace through this nested condition program
5.05 Create a repeat-until loop that counts down from sixty to zero
5.06 Create a list of five favorite foods
5.07 Create a bullet system where a spaceship sprite clones a bullet when space is…
5.08 Draw a regular hexagon using pen and motion blocks
5.09 Create a program that plays a five-note ascending scale
5.10 Test your game with a classmate
5.11 Explain the difference between AI and regular programs
5.12 Present your capstone game to the class

Key terms: Scratch, Stage, Sprite, Block, Variable, Scope, Assignment, Data Type, If, Else

I still need help with:

Chapter Map — Digital Citizenship

- 6.01 Online Identity
- 6.02 Privacy and Data
- 6.03 Cyberbullying Prevention
- 6.04 Digital Footprint Advanced
- 6.05 Online Reputation
- 6.06 Copyright Basics
- 6.07 Creative Commons
- 6.08 Fake News Detection
- 6.09 Screen Time Balance
- 6.10 Digital Wellness
- 6.11 Responsible Posting
- 6.12 Digital Citizenship Pledge

Key terms to know:

Lesson 6.01 Online Identity

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand the concept of online identity and digital persona] [- I can compare online identity with real-world identity] [- I can create a safe and responsible online profile]

Key Words

Identity	How you present yourself online with username and profile picture together
Profile	Page showing your information like name photo and interests to others
Username	Name you choose to log in and be known online
Password	Secret code proving you are owner of your account safely

Core Idea

Online identity is how you present yourself on the Internet. It includes your username, profile picture, bio, and content you share.

Sketch the diagram or concept here.

Worked Example

Step 1: Students create safe online identities.
Step 2: They practice sharing personality without compromising privacy.

Try It (write your answers)

1. Explain: Understand the concept of online identity and digital persona.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Online Identity.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

_____

I still need help with:

_____

Lesson 6.02 Privacy and Data

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand what personal data is and why it needs protection] [- I can identify different types of data collected by apps and websites] [- I can apply privacy settings to protect personal information]

Key Words

Privacy	Right to keep personal information away from strangers online always securely
Data	Personal facts like address or photo stored about you on internet
Consent	Permission you give before someone collects or uses your data legally
Settings	Options controlling who can see your information and posts online

Core Idea

Personal data includes your name, age, address, phone number, email, photos, location, contacts, and browsing history.

Sketch the diagram or concept here.

Worked Example

Step 1: Students practice real-world privacy analysis.
Step 2: They develop critical evaluation skills for digital privacy.

Try It (write your answers)

1. Explain: Understand what personal data is and why it needs protection.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Privacy and Data.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.03 Cyberbullying Prevention

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can define cyberbullying and recognize its different forms] [- I can understand the impact of cyberbullying on vICTims] [- I can learn strategies to prevent and respond to cyberbullying]

Key Words

Cyberbullying	Hurtful messages or posts targeting someone repeatedly online without stop
Block	Preventing bully from contacting you by cutting connection completely instantly
Report	Telling platform or trusted adult about harmful behaviour for help
Evidence	Saved screenshots proving bullying happened for showing to adults later

Core Idea

Cyberbullying is using technology to hurt, embarrass, or threaten another person repeatedly.

Sketch the diagram or concept here.

Worked Example

Step 1: Students create educational materials for their school community.
Step 2: They apply their knowledge to help others.

Try It (write your answers)

1. In your own words, what is cyberbullying?

Answer: Write a short definition using what you learned today.

2. Give one real-life example of Cyberbullying Prevention.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

_____

I still need help with:

_____

Lesson 6.04 Digital Footprint Advanced

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand how digital footprints are created and stored] [- I can define the long-term consequences of online actions] [- I can take steps to manage and reduce your digital footprint]

Key Words

Digital Footprint	Trail of data you leave whenever you use internet
Active Footprint	Data you share deliberately like posting photos online
Passive Footprint	Data collected secretly like cookies tracking your browsing
Reputation	What people think about you based on your online actions

Core Idea

A digital footprint is the trail of data you leave when using the Internet.

Sketch the diagram or concept here.

Worked Example

Step 1: Students reflect on their digital history and plan their digital future.
Step 2: They understand that they are actively creating their permanent record.

Try It (write your answers)

1. Explain: Understand how digital footprints are created and stored.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Digital Footprint Advanced.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.05 Online Reputation

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand how online reputation affects opportunities] [- I can build a positive digital presence through intentional content] [- I can recover from reputation damage through corrective actions]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Online reputation is how others perceive you based on your digital presence.

Sketch the diagram or concept here.

Worked Example

Step 1: students take concrete steps to improve their digital reputation.
Step 2: They practice proactive reputation management.

Try It (write your answers)

1. Explain: Understand how online reputation affects opportunities.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Online Reputation.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.06 Copyright Basics

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand what copyright protects and how it works] [- I can identify copyrightable materials and public domAIn content] [- I can apply fAIr use principles for educational purposes]

Key Words

Copyright	Legal right of creator to control copying of their work for years
Original	Work created by you that has not been copied from elsewhere truthfully
Permission	Approval needed before using someone else's creation legally without stealing
Fair Use	Limited use allowed for education without asking permission legally sometimes

Core Idea

Copyright is a legal right that gives creators control over their original works. It automatically applies when a work is created in a fixed form.

Sketch the diagram or concept here.

Worked Example

Step 1: students create practical copyright guides for their peers.
Step 2: They apply copyright knowledge to educational context.

Try It (write your answers)

1. Explain: Understand what copyright protects and how it works.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Copyright Basics.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.07 Creative Commons

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand Creative Commons licenses and their purposes] [- I can identify different CC license types and their conditions] [- I can find and use Creative Commons content legally in projects]

Key Words

Creative Commons	License letting creators share work with clear reuse rules for everyone
Attribution	Giving credit to original author when you reuse their work respectfully
ShareAlike	Rule requiring you share new work under same free license
License	Document stating what others may or may not do with your work

Core Idea

Creative Commons or CC provides free licenses that let creators share work with specific permissions. CC BY requires giving credit to the creator.

Sketch the diagram or concept here.

Worked Example

Step 1: students practice finding and using CC content legally.
Step 2: They build attribution habits for ethical content use.

Try It (write your answers)

1. Explain: Understand Creative Commons licenses and their purposes.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Creative Commons.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.08 Fake News Detection

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify characteristics of fake news and misinformation] [- I can apply verification techniques to evaluate news sources] [- I can develop critical thinking skills for consuming online information]

Key Words

Fake News	False information deliberately created to mislead readers quickly online for clicks
Source	Place where information originally comes from and must be checked
Verify	Checking facts with trusted sources before believing or sharing anything quickly
Bias	Unfair leaning toward one side that distorts truth in reporting often

Core Idea

Fake news is false information presented as real news.

Sketch the diagram or concept here.

Worked Example

Step 1: students build practical verification tools.
Step 2: They apply systematic checking to real information evaluation.

Try It (write your answers)

1. List or name: characteristics of fake news and misinformation.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Fake News Detection.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.09 Screen Time Balance

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand the effects of excessive screen time on health] [- I can create a balanced schedule for screen and non-screen activities] [- I can apply strategies for healthy technology use habits]

Key Words

Screen Time	Hours spent looking at phone computer or TV each day together
Balance	Dividing time equally between online and offline activities for health daily
Break	Short pause away from screen to rest eyes and mind peacefully
Wellness	Overall health including eyes sleep and mood affected by screen use

Core Idea

Excessive screen time can cause eye strain, headaches, poor posture, sleep disruption from blue light, reduced physical activity, and decreased attent

Sketch the diagram or concept here.

Worked Example

Step 1: students create practical schedules that balance technology use with healthy habits.
Step 2: They apply health knowledge to personal planning.

Try It (write your answers)

1. Explain: Understand the effects of excessive screen time on health.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Screen Time Balance.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.10 Digital Wellness

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can recognize signs of digital stress and technology addICTion] [- I can develop healthy relationships with technology] [- I can practice mindfulness in digital interactions]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Digital wellness means maintaining a healthy relationship with technology.

Sketch the diagram or concept here.

Worked Example

Step 1: students create community wellness initiatives.
Step 2: They practice promoting healthy habits among peers.

Try It (write your answers)

1. Explain: Recognize signs of digital stress and technology addiction.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Digital Wellness.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.11 Responsible Posting

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand the impact of online posts on others] [- I can apply the think-before-you-post framework] [- I can create positive and constructive online contributions]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Every online post affects others and your reputation. Before posting use the THINK framework. Is it True? Do I need to say it? Is it Helpful?

Sketch the diagram or concept here.

Worked Example

Step 1: students honestly evaluate their own posting habits.
Step 2: They create personal guidelines for responsible online communication.

Try It (write your answers)

1. Explain: Understand the impact of online posts on others.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Responsible Posting.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.12 Digital Citizenship Pledge

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can synthesize all digital citizenship concepts into personal commitments] [- I can create a comprehensive digital citizenship pledge] [- I can commit to responsible technology use and advocate for others]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

A digital citizenship pledge is a personal commitment to behave responsibly and ethically online.

Sketch the diagram or concept here.

Worked Example

Step 1: students create formal pledges demonstrating their complete understanding.
Step 2: They commit to specific behaviors they will follow.

Try It (write your answers)

1. Explain: Synthesize all digital citizenship concepts into personal commitments.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Digital Citizenship Pledge.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Quick Revision — Digital Citizenship

6.01 Evaluate these online identities for safety and appropriateness
6.02 Classify each data type as sensitive or non-sensitive
6.03 Respond to these situations
6.04 Define this person's digital footprint
6.05 Evaluate these online behaviors for reputation impact
6.06 Is this copyright infringement
6.07 Match each Creative Commons license to its requirements
6.08 Evaluate this news story
6.09 Rate your current screen habits as healthy or unhealthy for each category
6.10 Identify three signs of unhealthy technology use in your own habits
6.11 Evaluate these posts using the THINK framework
6.12 Present your digital citizenship pledge to the class

Key terms: Identity, Profile, Username, Password, Privacy, Data, Consent, Settings, Cyberbullying, Block

I still need help with:

Glossary

AI	Machines that learn and make decisions like humans without explicit programming rules
AVERAGE	Function dividing total by count to find middle value for group
Access Point	Device letting wireless devices join wired network through Wi-Fi signals freely
Active Footprint	Data you share deliberately like posting photos online
Alignment	Positioning text to left centre or right of page neatly
Animation	Creating movement by showing many pictures very quickly together
Append	Adding new item to end of existing list quickly without rebuilding
Apply	Using what you learned to solve new problem or daily task usefully
Ascending	Order from smallest to largest or A to Z upwards progressively
Assignment	Storing a value into variable using equals sign for later use
Attribution	Giving credit to original author when you reuse their work respectfully
Audio	Sound or music recorded and played through computer speakers
Axis	Line along side or bottom showing measurement scale clearly
BCC	Sending hidden copy where others cannot see recipient
Background	Bottom layer that holds main colour or photo behind everything
Backup	Copy of files kept safely to prevent loss from damage
Balance	Dividing time equally between online and offline activities for health daily
Bandwidth	Maximum data that can travel per second across a connection steadily
Bias	Unfair leaning toward one side that distorts truth in reporting often
Block	Colourful puzzle piece representing one programming instruction visually
Blur	Softening image so details look less sharp and smooth
Branch	Separate path program can take depending on different condition results possible
Break	Command stopping loop immediately even if condition still true for exit
Brightness	How light or dark the whole picture appears to eyes
Bus	All devices sharing one single cable like passengers on same road
CC	Sending same email copy to another person visibly
Canvas	Central area where your picture appears for editing directly with tools
Cell	Single box at intersection of row and column holding data
Chart	Picture like bar or pie showing data visually for understanding
Clear	Removing all pen drawings from stage to start fresh again
Clear Filter	Removing filter to show all rows again as before
Clip	Short piece of video you can move or trim separately
Clone	Copying clean area to cover unwanted spot seamlessly without trace
Cloud Storage	Saving files online so you access them from anywhere
Concept	Main idea that helps you understand the lesson topic clearly and correctly
Condition	Question answerable as true or false deciding which path program takes
Conditional Formatting	Automatically colouring cells that meet a rule like marks below 35
Consent	Permission you give before someone collects or uses your data legally
Contrast	Difference between darkest and lightest parts of image for clarity
Control	Block deciding when and how many clones appear on stage
Copyright	Legal right of creator to control copying of their work for years
Create Clone	Making new copy of sprite during program run with code
Creative Commons	License letting creators share work with clear reuse rules for everyone
Criteria	Rule deciding which rows stay visible after filtering operation for accuracy
Crop	Cutting outer parts to keep only important area of photo
Cyberbullying	Hurtful messages or posts targeting someone repeatedly online without stop
DNS	System translating website names into number addresses automatically for computers
Data	Facts like numbers and words stored on a computer
Data Point	Single value plotted as dot or bar inside chart
Data Type	Kind of value like number or text deciding what operations allowed

Dataset	Collection of related data stored together in one place
Delete Clone	Removing extra copies when no longer needed to save memory
Descending	Order from largest to smallest or Z to A downwards progressively
Digital Footprint	Trail of data you leave whenever you use internet
Dock	Side area where layers and brushes panels stay neatly organised together
Dropdown	Small menu appearing when you click filter arrow for choices
Dropdown List	Menu limiting choice to approved options like Yes or No
Editing	Changing picture to improve look or fix mistakes digitally with tools
Effect	Special style applied to make picture more attractive quickly
Element	Single value stored at one position inside a list for retrieval
Elif	Extra check when first condition fails but another might succeed sequentially
Else	Alternative block running when if condition proves false for completeness
Encryption	Scrambling data so only authorised person can read it with key
Episode	Single audio story that is part of larger series
Equals	Sign at formula start showing calculation should happen immediately after typing
Error Alert	Message warning you entered wrong type of data for cell
Evidence	Saved screenshots proving bullying happened for showing to adults later
Example	Specific case showing how the concept works in real life situations
Extension	Extra feature adding new blocks like pen or music to Scratch
FTP	Rule for uploading or downloading files between computers efficiently and safely
Fair Use	Limited use allowed for education without asking permission legally sometimes
Fake News	False information deliberately created to mislead readers quickly online for clicks
Feather	Softening edge of selection so change looks natural and blended
Filter	One-click effect changing colours or brightness of whole picture instantly
Firewall	Barrier blocking unwanted traffic from entering network without permission for safety
Flow	Order in which program checks conditions from outer to inner sequentially
Font	Design style of letters like Arial or Noto Sans for reading
For Loop	Repeating fixed number of times using counter variable automatically for control
Foreground	Top layer that appears in front of all other layers
Formula	Equation starting with equals calculating result from other cells automatically
Forward	Sending received email to another person with original content
Frame	Single still picture that is part of a longer video
GIMP	Free program for advanced photo editing like Photoshop without cost
HTTP	Rule for fetching web pages from server to your browser quickly
Heading	Big bold words at top that catch attention quickly
Heal	Blending copied area so repair looks natural and invisible to eyes
Highlight	Colouring cell background to draw attention to important values quickly
Host	Person who speaks and guides the podcast for listeners
Hotspot	Area where Wi-Fi is available for phones and laptops to connect
Hub	Simple device broadcasting data to all connected devices at once together
Hue	Name of colour like red blue or green on colour wheel
IP Address	Unique numbers identifying each device on network like house address
ISP	Company giving internet connection to homes and schools for monthly fee
Identity	How you present yourself online with username and profile picture together
If	Decision checking condition and running code only when true otherwise skipping
Index	Position number starting at zero showing where item sits in list
Information	Useful meaning produced after processing raw data carefully
Integration	Combining different media types into one smooth presentation for audience
Interactive	Letting viewer click or choose what happens next inside project

Key	Column chosen as basis for deciding sorted order of rows
LAN	Network covering one building like school lab or office
Lasso	Freehand tool for drawing selection around irregular shapes with mouse carefully
Layer	Transparent sheet stacked to build complex picture step by step
Layout	Arrangement of text and pictures to look balanced neatly
Legend	Key explaining what each colour or symbol means in chart
License	Document stating what others may or may not do with your work
List	Ordered collection storing many values together under one name like marks
Logic	Reasoning combining many conditions using and or or correctly together
Loop	Structure repeating block of code many times without rewriting same lines
MAX	Function finding largest number in selected range instantly for comparison
MIN	Function finding smallest number in selected range instantly for comparison
Machine Learning	Training computers by showing many examples instead of writing rules
Media	Single form like audio or video used inside larger presentation
Microphone	Device that captures your voice and sends to computer
Motion	Change of position of object from one place another
Multimedia	Combination of text sound images and video together to communicate
Narrative	Way you organise story with beginning middle and end
Nested If	If statement placed inside another if for more detailed decision making
Network	Group of devices connected to share information and resources together
Opacity	How see-through or solid a layer appears on screen
Operator	Symbol like plus or minus telling what math to do
Original	Work created by you that has not been copied from elsewhere truthfully
PAN	Tiny network around one person like phone to watch
Packet	Small piece of message travelling independently to be reassembled later correctly
Passive Footprint	Data collected secretly like cookies tracking your browsing
Password	Secret word protecting network from unknown users trying to connect
Patch	Replacing damaged part with good texture from nearby area smoothly
Pen	Extension allowing sprite to draw lines as it moves across stage
Permission	Approval needed before using someone else's creation legally without stealing
Pivot Table	Summarising large data quickly by grouping and counting with drag
Playback	Listening again to recorded sound through speakers clearly
Podcast	Series of audio episodes you can listen anytime online
Poster	Large design combining text and images to share message
Practice	Repeating task many times to improve skill and remember better always
Prediction	Guess made by trained model about new unseen data based on learning
Privacy	Right to keep personal information away from strangers online always securely
Profile	Page showing your information like name photo and interests to others
Protocol	Agreed rule letting different computers talk to each other correctly without confusion
Range	Group of selected cells like A1 to C5 for calculation together
Raw Data	Unprocessed facts collected before cleaning or organising properly
Record	Saving your voice or sound as a digital file
Reference	Address like B2 pointing to value in another cell for calculation
Reply All	Answering all people included in original email together
Report	Telling platform or trusted adult about harmful behaviour for help
Reputation	What people think about you based on your online actions
Resolution	Number of dots per inch deciding how sharp image looks when printed
Retouch	Fixing small flaws like spots or red eyes in photo digitally
Ring	Devices linked in circle where data travels around sequentially one by one
Router	Box directing data between different networks efficiently without delay for speed
Row Label	Category appearing along left side of pivot table for grouping

Rule	Condition like greater than 80 that decides which cells change colour
SMTP	Rule for sending emails from your device to recipient's server reliably
SUM	Function adding all numbers in selected range quickly without manual typing
Saturation	Intensity of colours from dull grey to vivid bright appearance
Scale	Graduated colours showing low to high values across range visually nicely
Scene	Single part of story showing one moment or place
Scope	Part of program where variable can be seen and used correctly
Scratch	Visual programming tool using blocks to create stories and games
Screen Time	Hours spent looking at phone computer or TV each day together
Selection	Dotted line marking area you want to edit separately from rest
Service	Useful facility provided through internet like email or video call
Settings	Options controlling who can see your information and posts online
ShareAlike	Rule requiring you share new work under same free license
Shared Drive	Folder where many people view and edit same files
Signal	Radio wave carrying data through air between devices invisibly around us
Size	How big or small the text appears on page or screen
Slide	Single page inside presentation showing text and images together
Sort	Putting rows in alphabetical or numerical order like A-Z quickly
Source	Place where information originally comes from and must be checked
Sprite	Character or object that moves and acts on the stage
Stage	Area where your characters move and show their actions live
Stamp	Copying sprite image permanently onto stage background like printing once
Star	All devices connected to one central hub like spokes of wheel
Story	Sequence of events told with words pictures and sound together
Streaming	Playing video or music directly without downloading whole file completely first
Subscribe	Following podcast to receive new episodes automatically when released
Switch	Box connecting many devices inside one network and forwarding data correctly fast
Sync	Keeping same files updated across all your devices automatically
TCP	Rule ensuring data arrives correctly and in order without missing pieces
Template	Ready-made design that keeps all slides looking consistent neatly
Timeline	Bar showing order and length of clips in your project
Toolbox	Panel holding brushes selection and paint tools together on side
Topology	Shape showing how devices are arranged and connected in network physically
Training	Feeding data to model so it learns patterns and improves accuracy gradually
Transition	Effect that moves you smoothly from one slide next
Typography	Art of arranging text to look beautiful and readable together
Username	Name you choose to log in and be known online
Validation	Checking entered data matches allowed type or range before accepting
Value	Number being counted summed or averaged inside pivot table results
Variable	Named container storing value that program can change later dynamically
Verify	Checking facts with trusted sources before believing or sharing anything quickly
Video	Moving pictures made from many still images played quickly
Virus	Harmful program that copies itself and damages files or steals information
Voiceover	Recorded speech added over pictures to explain story clearly
WAN	Huge network spanning cities or countries like internet globally
Wand	One-click tool selecting areas of similar colour automatically for speed
Wellness	Overall health including eyes sleep and mood affected by screen use
While Loop	Repeating while condition stays true checking each time before running
Wi-Fi	Radio technology connecting devices without cables over short distance conveniently
Worksheet	Single page of spreadsheet containing grid of cells for work

Self-Assessment

Chapter	I Can	With Help	Not Yet
Networking and Internet	☐	☐	☐
Advanced Graphics	☐	☐	☐
Multimedia	☐	☐	☐
Data Handling	☐	☐	☐
Programming with Scratch	☐	☐	☐
Digital Citizenship	☐	☐	☐